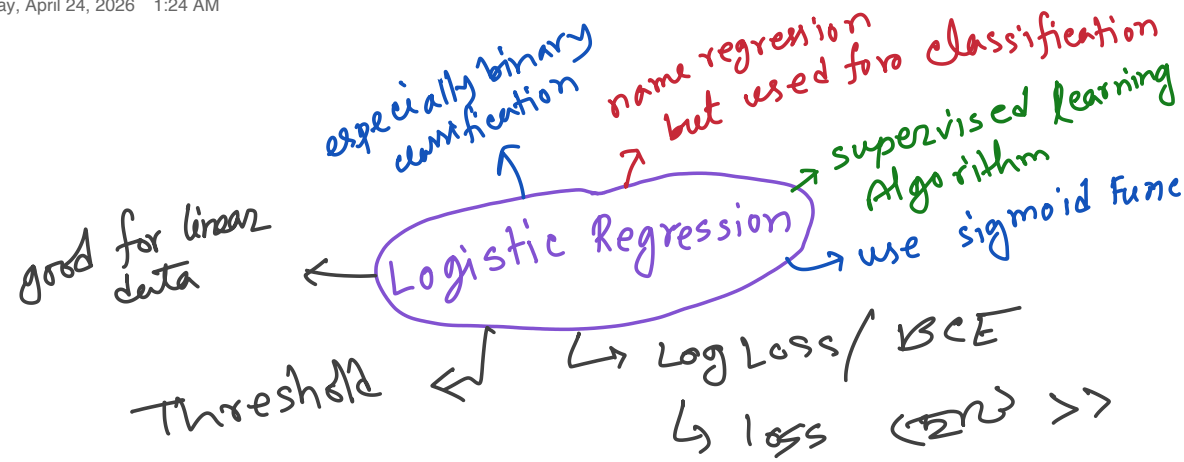


Logistic Regression

Friday, April 24, 2026 1:24 AM



Working Principle:

① linear equation:

$$z = wx + b$$

weight bias

② use Sigmoid func:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

→ output → between 0 & 1.

probability =

Patient-A → 0.20 → 0

Patient-B → 0.48 → 0

" - C → 0.90 → 1

be2 Threshold = 0.5

Model selection depends on data characteristics. For linear data we use Logistic Regression or Linear SVM. For non-linear structured data we use SVM with kernel or Decision Tree. For noisy data we prefer simpler models. For large datasets SGDClassifier is efficient, and for interpretability Decision Tree is preferred.

